

I. Graphs of Tangent, Cotangent, Secant, and Cosecant

Periodic Properties:

The functions **tangent** and **cotangent** have period π .

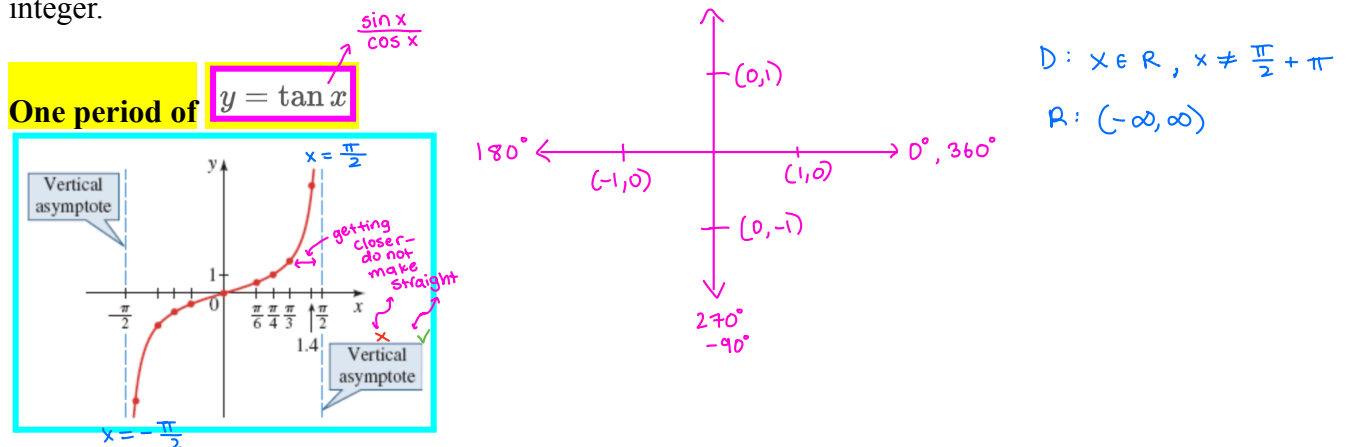
$$\tan(x + \pi) = \tan x \qquad \cot(x + \pi) = \cot x$$

The functions **cosecant** and **secant** have period 2π .

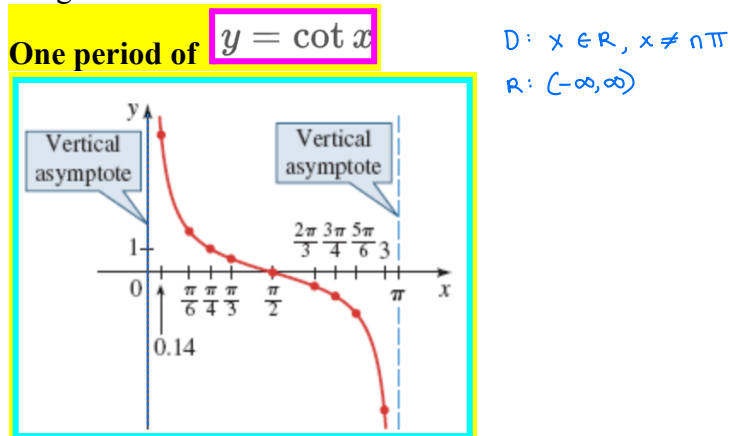
$$\csc(x + 2\pi) = \csc x \qquad \sec(x + 2\pi) = \sec x$$

TANGENT

The graph of $y = \tan x$ approaches the vertical lines $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$, so these lines are **vertical asymptotes**. The vertical asymptotes of $y = \tan x$ are $x = \frac{\pi}{2} + n\pi$ where n is an integer.

**COTANGENT**

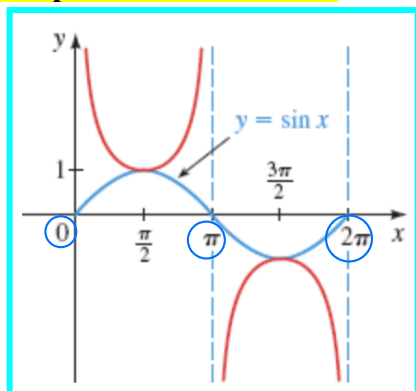
The graph of $y = \cot x$ is undefined for $x = n\pi$, with n an integer, so its graph has vertical asymptotes at these values. The vertical asymptotes of $y = \cot x$ are $x = n\pi$ where n is an integer.



COSECANT

To graph the cosecant we use the identity $\csc x = \frac{1}{\sin x}$. The graph of $y = \csc x$ has vertical asymptotes at $x = n\pi$, for n an integer. $\sin x = 0$

One period of $y = \csc x$



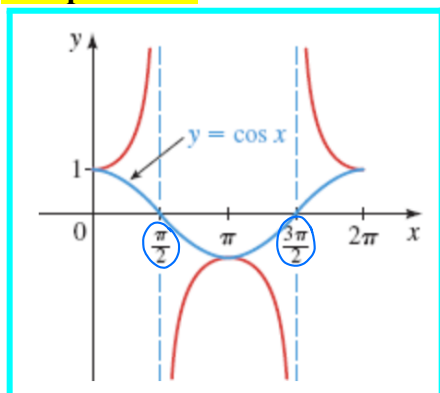
$$D: x \in \mathbb{R}, x \neq n\pi$$

$$R: (-\infty, -1] \cup [1, \infty)$$

SECANT

To graph the secant we use the identity $\sec x = \frac{1}{\cos x}$. The graph of $y = \sec x$ has vertical asymptotes at $x = (\pi/2) + n\pi$, for n an integer. $\cos x = 0$

One period of $y = \sec x$



$$D: x \in \mathbb{R}, x \neq \frac{\pi}{2} + n\pi$$

$$R: (-\infty, -1] \cup [1, \infty)$$

Graphs of Transformations of Tangent and Cotangent

The functions $y = a \tan kx$ and $y = a \cot kx$, ($k > 0$), have period $\frac{\pi}{k}$.

To graph one period of $y = a \tan kx$, an appropriate interval is $\left(-\frac{\pi}{2k}, \frac{\pi}{2k}\right)$.

To graph one period of $y = a \tan k(x - b)$, an appropriate interval is $\left(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b\right)$.

To graph one period of $y = a \cot kx$, an appropriate interval is $\left(0, \frac{\pi}{k}\right)$.

To graph one period of $y = a \cot k(x - b)$, an appropriate interval is $\left(0 + b, \frac{\pi}{k} + b\right)$.

II. Graphs of Transformations of Cosecant and Secant

The functions $y = a \csc kx$ and $y = a \sec kx$, ($k > 0$), have period $\frac{2\pi}{k}$.

An appropriate interval on which to graph one complete period is $[0, 2\pi/k]$.

To graph one period of $y = a \csc k(x - b)$ or $y = a \sec k(x - b)$, an appropriate interval is

$$\left(0 + b, \frac{2\pi}{k} + b\right)$$

Example 1: Find the period, and graph the function.

a. $y = 4\tan(4x - 2\pi)$

$$4x - 2\pi = -\frac{\pi}{2}$$

$$4x = \frac{3\pi}{2}$$

$$x = \frac{3\pi}{8}$$

$$4x - 2\pi = \frac{\pi}{2}$$

$$4x = \frac{5\pi}{2}$$

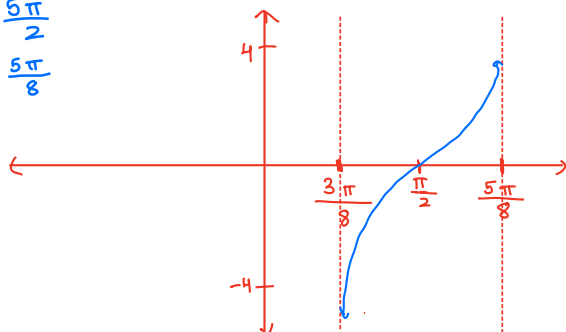
$$x = \frac{5\pi}{8}$$

$$y = 4\tan\left(4\left(x - \frac{\pi}{2}\right)\right)$$

$$P = \frac{\pi}{k} = \frac{\pi}{4}, I = \left(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b\right)$$

$$= \left(-\frac{\pi}{8} + \frac{\pi}{2}, \frac{\pi}{8} + \frac{\pi}{2}\right)$$

$$= \left(\frac{3\pi}{8}, \frac{5\pi}{8}\right)$$



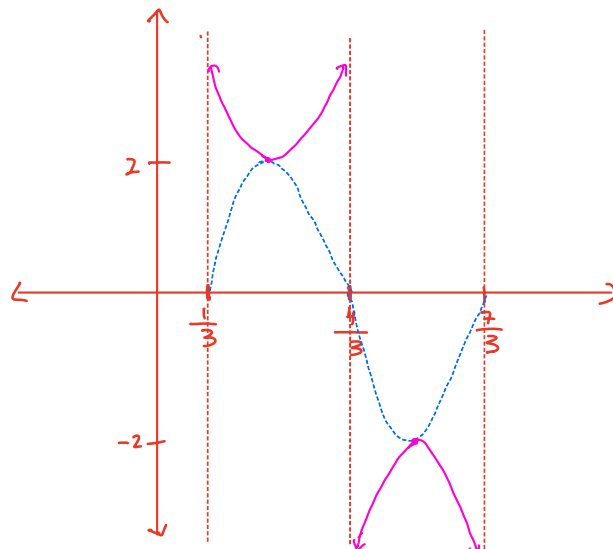
b. $y = 2\csc\left(\pi x - \frac{\pi}{3}\right)$

$$y = 2\csc\left(x - \frac{1}{3}\right)$$

$$P = \frac{2\pi}{\pi} = 2, I: \left(0 + b, \frac{2\pi}{\pi} + b\right)$$

$$: \left(0 + \frac{1}{3}, 2 + \frac{1}{3}\right)$$

$$\left(\frac{1}{3}, \frac{7}{3}\right)$$

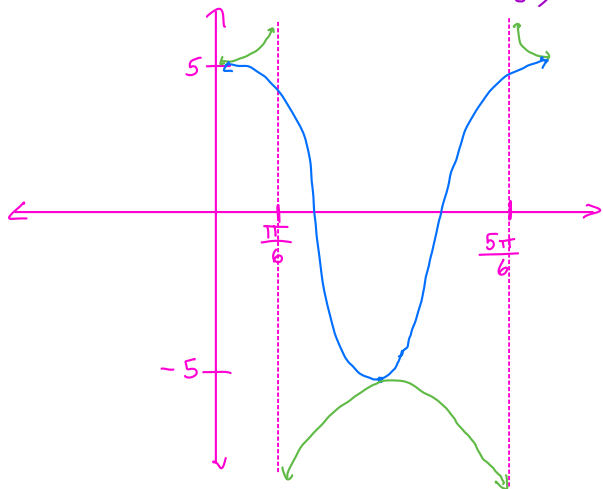


c. $y = 5\sec\left(3x - \frac{\pi}{2}\right)$

$$= 5\sec\left(3\left(x - \frac{\pi}{6}\right)\right)$$

$$P = \frac{2\pi}{k} = \frac{2\pi}{3}, I: \left(0 + \frac{\pi}{6}, \frac{\pi}{6} + \frac{2\pi}{3}\right)$$

$$\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$$

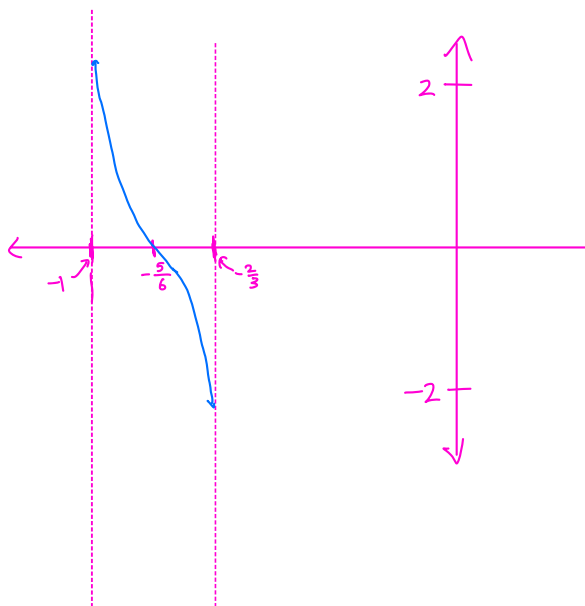


d. $y = 2\cot(3\pi x + 3\pi)$

$$= 2\cot(3\pi(x + 1))$$

$$P = \frac{\pi}{k} = \frac{\pi}{3\pi} = \frac{1}{3}, I: \left(0 + (-1), \frac{\pi}{3\pi} + (-1)\right)$$

$$\left(-1, -\frac{2}{3}\right)$$



$$\cos^{-1} x : [0, \pi]$$

QI QII
⊕ ⊖

$$\sin^{-1}(x) : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

QIV, QI
⊖ ⊕

$$\tan^{-1}(x) : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

QIV, QI
⊖ ⊕

I. The Inverse Sine Function

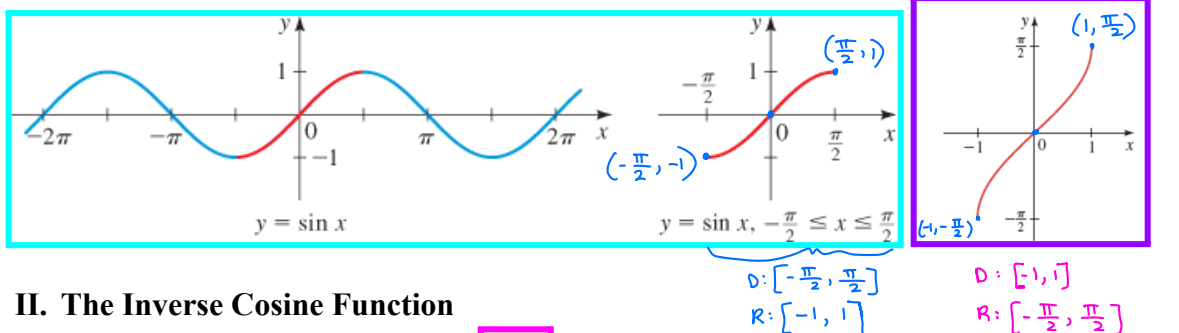
The **inverse sine function** is the function $\sin^{-1}$ with domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$ defined by $\sin^{-1} x = y \Leftrightarrow \sin y = x$. The inverse sine function is also called arcsine, denoted by arcsin.

$y = \sin^{-1} x$ is the number in the interval $[-\pi/2, \pi/2]$ whose sine is x .

The inverse sine function has these **cancellation properties**:

$$\sin(\sin^{-1} x) = x \text{ for } -1 \leq x \leq 1$$

$$\sin^{-1}(\sin x) = x \text{ for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$



II. The Inverse Cosine Function

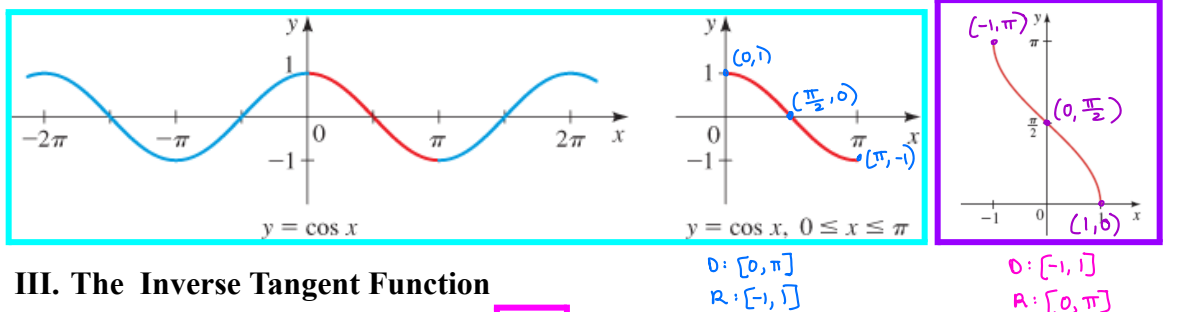
The **inverse cosine function** is the function $\cos^{-1}$ with domain $[-1, 1]$ and range $[0, \pi]$ defined by $\cos^{-1} x = y \Leftrightarrow \cos y = x$. The inverse cosine function is also called arccosine, denoted by arccos.

$y = \cos^{-1} x$ is the number in the interval $[0, \pi]$ whose cosine is x .

The inverse cosine function has these **cancellation properties**:

$$\cos(\cos^{-1} x) = x \text{ for } -1 \leq x \leq 1$$

$$\cos^{-1}(\cos x) = x \text{ for } 0 \leq x \leq \pi$$



III. The Inverse Tangent Function

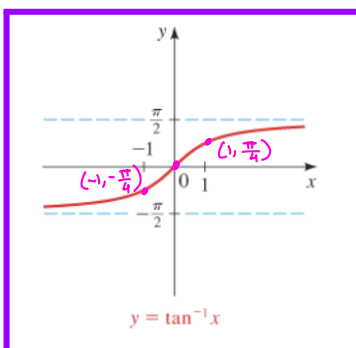
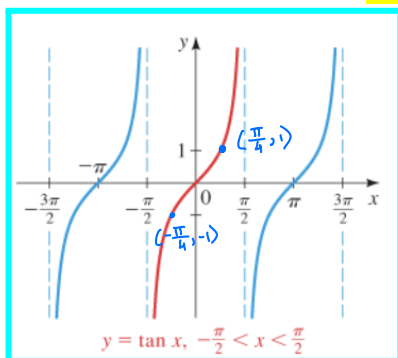
The **inverse tangent function** is the function $\tan^{-1}$ with domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$ defined by $\tan^{-1} x = y \Leftrightarrow \tan y = x$. The inverse tangent function is also called arctangent, denoted by arctan.

$y = \tan^{-1} x$ is the number in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ whose tangent is x .

The inverse tangent function has these **cancellation properties**:

$$\tan(\tan^{-1}x) = x \text{ for } x \in \mathbb{R}$$

$$\tan^{-1}(\tan x) = x \text{ for } -\frac{\pi}{2} < x < \frac{\pi}{2}$$

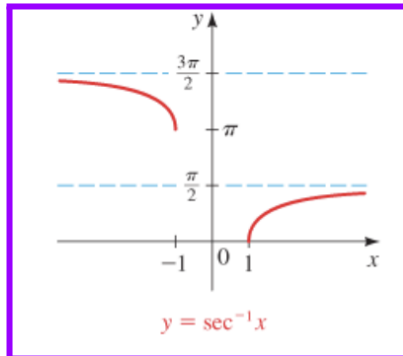
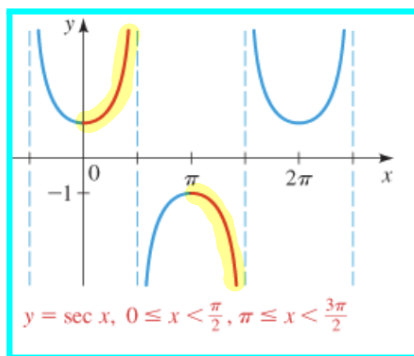


$$y = \tan x \rightarrow \begin{matrix} D: (-\frac{\pi}{2}, \frac{\pi}{2}) \\ R: (-\infty, \infty) \end{matrix}$$

$$y = \tan^{-1}x \rightarrow \begin{matrix} D: (-\infty, \infty) \\ R: (-\frac{\pi}{2}, \frac{\pi}{2}) \end{matrix}$$

The Inverse Secant, Cosecant, and Cotangent Functions

To define the inverse functions of the secant, cosecant, and cotangent functions, we restrict the domain of each function to a set on which it is one-to-one and on which it attains all its values.



$$\begin{matrix} Q I & + & Q III \\ \oplus & & \ominus \end{matrix}$$

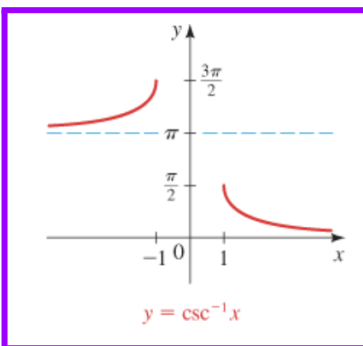
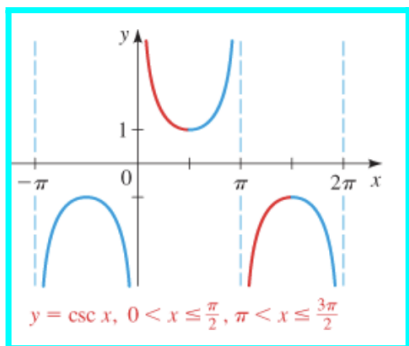
$$D: [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$$

$$R: (-\infty, -1] \cup [1, \infty)$$

$$y = \sec^{-1}x$$

$$D: (-\infty, -1] \cup [1, \infty)$$

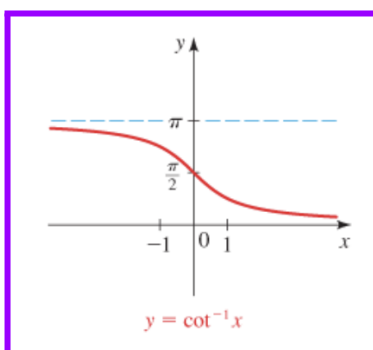
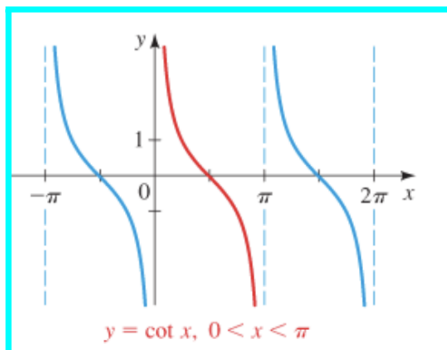
$$R: [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$$



$$\begin{matrix} Q I & & Q III \\ \oplus & & \ominus \end{matrix}$$

$$y = \csc x: \begin{matrix} D: (0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}] \\ R: (-\infty, -1] \cup [1, \infty) \end{matrix}$$

$$y = \csc^{-1}x: \begin{matrix} D: (-\infty, -1] \cup [1, \infty) \\ R: (0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}] \end{matrix}$$



$$\begin{matrix} Q I & & Q II \\ \oplus & & \ominus \end{matrix}$$

$$y = \cot x$$

$$D: (0, \pi)$$

$$R: (-\infty, \infty)$$

$$y = \cot^{-1}x$$

$$D: (-\infty, \infty)$$

$$R: (0, \pi)$$

Example 1: Find the exact value of each expression, if it is defined.

a. $\tan^{-1} 1$
 $\boxed{\frac{\pi}{4}}$

b. $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$
 $\boxed{\frac{5\pi}{6}}$

c. $\sin^{-1}(-1)$
 $\boxed{-\frac{\pi}{2}}$

Example 2: Use a calculator to find an approximate value of each expression correct to five decimal places, if it is defined.

a. $\sin^{-1}(0.13844)$
 $\boxed{\approx 0.13889}$

b. $\cos^{-1}\left(\frac{-3}{7}\right)$
 $\boxed{\approx 2.01371}$

Example 3: Find the exact value of the expression, if it is defined.

a. $\sin\left(\sin^{-1}\left(\frac{1}{4}\right)\right)$
 $\boxed{= \frac{1}{4}}$

b. $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$
 $\boxed{\frac{5\pi}{6}}$

c. $\tan^{-1}\left(\tan\left(\frac{2\pi}{3}\right)\right)$
 $\boxed{-\frac{\pi}{3}}$

d. $\sin(\sin^{-1}(5)) \neq [-1, 1]$
 No solution

e. $\sin(\sec^{-1}(-2))$
 $\sec \theta = -2$
 $\cos \theta = -\frac{1}{2}$
 $\theta = \frac{2\pi}{3}$
 $(-1)^2 + y^2 = (2)^2$
 $y = \sqrt{3}$
 $\sin(\sec^{-1}(-2)) = \frac{\sqrt{3}}{2}$

f. $\csc(\tan^{-1}(-\sqrt{3}))$
 $\tan \theta = -\sqrt{3}$
 $\theta = \frac{5\pi}{6}$
 $r^2 + (-\sqrt{3})^2 = r^2$
 $r = 2$
 $\csc(\tan^{-1}(-\sqrt{3})) = -\frac{2\sqrt{3}}{3}$

g. $\tan\left(\cos^{-1}\left(\frac{-\sqrt{2}}{2}\right)\right)$
 $\cos \theta = -\frac{\sqrt{2}}{2}$
 $\theta = \frac{3\pi}{4}$
 $(-\sqrt{2})^2 + y^2 = 2^2$
 $y = \sqrt{2}$
 $\tan(\cos^{-1}(-\frac{\sqrt{2}}{2})) = -1$

$\cos(\sin^{-1}\frac{1}{2})$
 $y = 1, r = 2$
 $x = \sqrt{4-1} = \sqrt{3}$
 $\cos \theta = \frac{\sqrt{3}}{2}$

Example 4: Rewrite the expression as an algebraic expression in x .

a. $\cos(\sin^{-1} x)$
 $\frac{x}{r} = \frac{y}{r}$
 $y = x, r = 1$
 $x = \sqrt{1-x^2}$
 $\cos \theta = \frac{x}{r} = \frac{\sqrt{1-x^2}}{1}$

b. $\sec(\tan^{-1} x)$
 $\frac{y}{x} = \frac{y}{x}, x = 1$
 $r = \sqrt{1+x^2}$
 $\sec \theta = \frac{r}{x} = \frac{\sqrt{1+x^2}}{1}$

$\sec(\tan^{-1}\frac{3}{5})$
 $y = 3, x = 5$
 $r = \sqrt{9+25}$
 $r = \sqrt{34}$
 $\sec \theta = \frac{r}{x} = \frac{\sqrt{34}}{5}$

Example 5: Find functions f and g such that $F = f \circ g$.

a. $F(x) = e^{\arcsin x}$
 $f(x) = e^x, g(x) = \arcsin x$
 $F = f(g(x))$
 $= f(\arcsin x)$
 $= e^{\arcsin x}$

b. $F(x) = (\tan^{-1} x)^2$
 $f(x) = x^2, g(x) = \tan^{-1} x$
 $F = f(g(x))$
 $= f(\tan^{-1} x)$
 $= (\tan^{-1} x)^2$